

gcloud Setup Guide

Audience: Ops engineers taking over platform management (GKE-based). **Scope:** authenticating and configuring gcloud on your machine. Requires tooling from **1. Tool Installation Guide** to already be installed. Connecting to the GKE cluster and all kubectl usage is covered in **3. Kubernetes Setup Guide**.

Prerequisites

- Tooling installed per **1. Tool Installation Guide**
 - A GCP identity (per the access matrix)
 - Access already granted per the platform access matrix
 - Target GCP Project ID(s) for dev/qa/prod
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Authentication

```
gcloud init
```

This opens a browser to authenticate (gcloud auth login under the hood) and lets you pick a default project interactively.

Headless/remote machine (no browser available):

```
gcloud auth login --no-launch-browser
```

Verify:

```
gcloud auth list    # your account should show, marked ACTIVE
```

Setting the Active Project

```
gcloud config set project PROJECT_ID
```

Since you'll be working across dev/qa/prod projects, it's worth knowing you can switch anytime without re-running init:

```
gcloud config set project OTHER_PROJECT_ID
```

Optionally set a default region/zone so you don't have to pass --region/--zone on every future command:

```
gcloud config set compute/region us-central1
gcloud config set compute/zone us-central1-a
```

Verify:

```
gcloud config list    # shows active account, project, region, zone
```

What's Next

At this point gcloud is installed, authenticated, and pointed at a project. Connecting kubectl to an actual GKE cluster (the get-credentials command, kubeconfig context renaming so kubectl shows

clean cluster names, and all day-to-day kubectl operations) is covered in **3. Kubernetes Setup Guide**.

Appendix: Testing This Guide on a Clean Machine

A fresh OS user account is enough — no gcloud config, no PATH entries, nothing left over from a real install:

- **macOS:** System Settings → Users & Groups → Add Account → new standard user → log in as them and run through the guide.
- **Windows:** Settings → Accounts → Family & other users → Add account → new local account → sign in and run through the guide.

If you need a genuinely separate OS install and can't download an ISO, browser/RDP cloud desktops sidestep the download restriction: - **Windows:** Azure Virtual Desktop, Windows 365 Cloud PC, AWS WorkSpaces - **macOS:** AWS EC2 Mac instances, MacStadium, MacinCloud

These are billed services — check with your team, and confirm whether device policy applies to them the same way it applies to ISOs.